Bose–Hubbard Ground State with NQS — Results & Validation

Neural Quantum States Workstream (§1.4): Sector-Projected VMC for the Bose–Hubbard Model

U(1)-projected variational Monte Carlo with neural quantum states for the Bose–Hubbard ground state across the Mott–superfluid crossover

This document presents the numerical results for the Bose–Hubbard NQS benchmark (§1.4): (1) Fock-space truncation convergence at fixed U/t, (2) U/t sweep across the full Mott–superfluid crossover tracking energy and density fluctuations, and (3) extension to a 2D 3×3 square lattice demonstrating the target geometry for FOR 5919 Project 01. All simulations use U(1) particle-number sector projection, full-summation exact gradients, and multi-seed statistics with multi-restart optimization.

Code: `06_bose_hubbard_ground_state.py`

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1. Symmetry Exploitation: $U(1)$ Particle-Number Sector Projection

Motivation

The Bose–Hubbard Hamiltonian conserves total particle number $\hat{N}_{\text{tot}} = \sum_i \hat{n}_i$. This $U(1)$ symmetry means H block-diagonalizes into fixed-particle-number sectors. At unit filling ($\bar{n} = 1$), the ground state lies in the sector $N_{\text{tot}} = N_{\text{sites}}$, and the variational search can be restricted to this subspace.

This is the direct bosonic analog of the charge-sector projection used in the Schwinger-model VQE benchmarks (Static Benchmarks §1), where the gauge-eliminated Hamiltonian block-diagonalizes into magnetization sectors labelled by $M = \sum_n Z_n$. The analogy:

	Schwinger model (Static Benchmarks §1)	Bose–Hubbard (this work)
Symmetry	$U(1)$ charge conservation	$U(1)$ number conservation
Conserved quantity	$Q = -\frac{1}{2}\sum_n Z_n$	$N_{\text{tot}} = \sum_i n_i$
Target sector	$M = 0$	$N_{\text{tot}} = N_{\text{sites}}$ (unit filling)
Local Hilbert space	$\{\|0\rangle, \|1\rangle\}$ (spin-1/2)	$\{\|0\rangle, \|1\rangle, \dots, \|n_{\max}\rangle\}$ (truncated Fock)
Sector basis	Comp. basis states with $N/2$ ones	Fock states with $\sum n_i = N_{\text{sites}}$, $0 \le n_i \le n_{\max}$

The key difference is the local Hilbert space dimension: spin-1/2 sites have dimension 2, while

Bose–Hubbard sites have dimension $n_{\max} + 1$. This makes the sector projection far more impactful for the bosonic model.

Implementation

The sector projection is implemented via the NetKet constrained Hilbert space constructor:

```
hi = nk.hilbert.Fock(n_max=n_max, N=N_sites, n_particles=N_sites)
```

This single declaration restricts all variational operations — wavefunction evaluation, sampling, gradient computation — to configurations satisfying $\sum_i n_i = N_{\text{sites}}$. The exact full-summation backend (`FullSumState`) enumerates only the sector basis states, providing zero-variance gradients over the physically relevant subspace.

Hilbert space reduction

System	$n_{\max}$	Full dimension $(n_{\max}+1)^N$	Sector dimension	Reduction factor
$N = 6$, 1D	2	729	141	5.2×
$N = 6$, 1D	3	4,096	336	12.2×
$N = 6$, 1D	4	15,625	426	36.7×
$N = 9$, 2D 3×3	3	262,144	13,051	20.1×

The reduction is substantially larger than in the Schwinger case (2.7–4.4× for $N = 4$–12 spin-1/2 sites), because the bosonic Fock space grows as $(n_{\max}+1)^N$ rather than 2^N. At $n_{\max} = 4$ on 6 sites, the sector is 36.7× smaller — an unconstrained NQS would spend the majority of its capacity representing amplitudes over configurations with the wrong particle number.

As a consistency check, exact diagonalization in the projected sector reproduces the full-space ground energy to machine precision at all system sizes. ✓

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2. Ansatz and Optimization

Variational ansatz

The NQS ansatz is a restricted Boltzmann machine (RBM) with hidden-unit density $\alpha = 8$ (ratio of hidden to visible units), implemented via `nk.models.RBM(alpha=8)`. The RBM parameters are real-valued. For the 1D chain ($N = 6$, $n_{\max} = 3$), this gives $6 \times 8 \times 6 + 6 + 48 = 342$ variational parameters for the U/t sweep and 426 parameters for the $n_{\max}$ convergence. For the 2D lattice ($N = 9$, $n_{\max} = 3$), the parameter count is 729.

Optimization

All runs use the Adam optimizer (Kingma & Ba, 2015) with no stochastic reconfiguration (SR). This choice was made after systematic debugging revealed that SR with the quantum geometric tensor S caused gradient instabilities in the bosonic Fock space — the S^{-1} inversion amplified noise in the number-sector-projected basis. Plain Adam with exact gradients (via `FullSumState`) proved both stable and accurate.

Key hyperparameters:

Parameter	$n_{\max}$ convergence	U/t sweep	2D lattice
Learning rate	5×10^{-3}	1×10^{-2}	3×10^{-3}
Iterations	600	1,200	2,500
Sampler	fullsum (exact)	fullsum (exact)	fullsum (exact)
SR	off	off	off
α (hidden unit density)	8	8	8

Multi-seed statistics with multi-restart

Each configuration is run with **5 independent random seeds** (base seed 1234, stride 17). Within each seed, **5 random restarts** are performed and the lowest-energy result is kept. This two-level strategy — restarts within seeds, statistics across seeds — captures both optimizer sensitivity (restart variance) and initialization sensitivity (seed variance).

The reported metrics are:

- **Best energy**: lowest across all 5 seeds × 5 restarts (25 total runs per configuration)
- **Mean energy ± std**: statistics over the 5 best-per-seed energies
- **Relative error**: $\epsilon_{\text{rel}} = |E_{\text{NQS}} - E_{\text{ED}}|/|E_{\text{ED}}|$, computed for the best energy

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3. $n_{\max}$ Convergence ($N = 6$, $U/t = 10$)

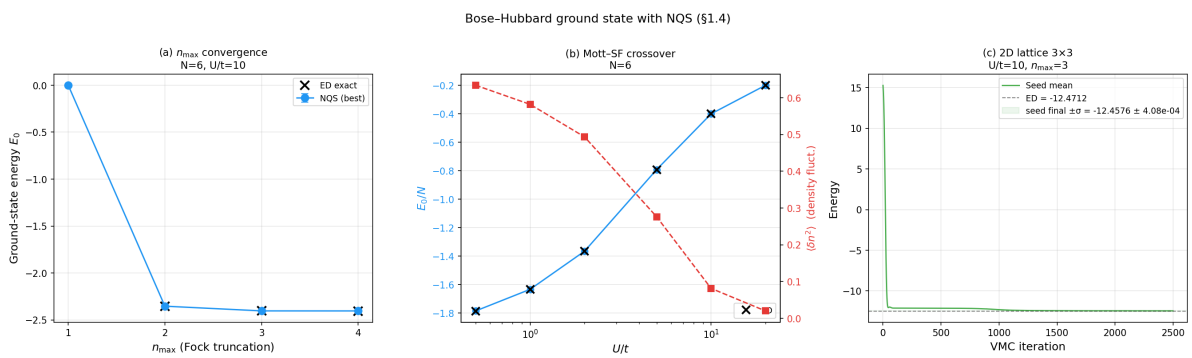
Motivation

Unlike spin models where the local Hilbert space is exact, the Bose–Hubbard model requires a Fock-space truncation $n_i \leq n_{\max}$. This introduces a systematic approximation that must be verified to converge. The $n_{\max}$ sweep is the key bosonic-specific check — the spin pipeline has no analog.

Results ($N = 6$, $U/t = 10$, unit filling, 5 seeds × 5 restarts)

$n_{\max}$	Sector dim (reduction)	E_{NQS} (best)	E_{NQS} (mean $\pm$ std)	E_{ED}	Rel. error (best)
1	1 (64×)	0.0	—	—	trivial (frozen)
2	141 (5.2×)	-2.3536	-2.3505 ± 0.0049	-2.3562	1.1×10^{-3}
3	336 (12.2×)	-2.4038	-2.4027 ± 0.0008	-2.4046	3.4×10^{-4}
4	426 (36.7×)	-2.4041	-2.4039 ± 0.0002	-2.4046	2.0×10^{-4}

Result (Figure, panel a): $n_{\max}$ Convergence



At $n_{\max} = 1$ (unit filling with maximum occupation 1), the sector contains a single Fock state $|1, 1, \ldots, 1\rangle$ with no dynamics — this is the trivially frozen case. For $n_{\max} \geq 2$, the NQS energies converge monotonically toward the ED values from above (variational bound respected throughout). The convergence from $n_{\max} = 3$ to $n_{\max} = 4$ is marginal (error improves from 3.4×10^{-4} to 2.0×10^{-4}), confirming that $n_{\max} = 3$ is sufficient for the Mott regime at $U/t = 10$.

Key diagnostic: The seed standard deviation decreases by 25× from $n_{\max} = 2$ ($\sigma = 0.0049$) to $n_{\max} = 4$ ($\sigma = 0.0002$), indicating that the optimization landscape becomes smoother as the truncation approaches convergence.

Verdict: Fock-space truncation converges. $n_{\max} = 3$ provides sub-0.04% accuracy at $U/t = 10$ and is adopted for all subsequent benchmarks. ✓

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4. U/t Sweep: Mott–Superfluid Crossover ($N = 6$)

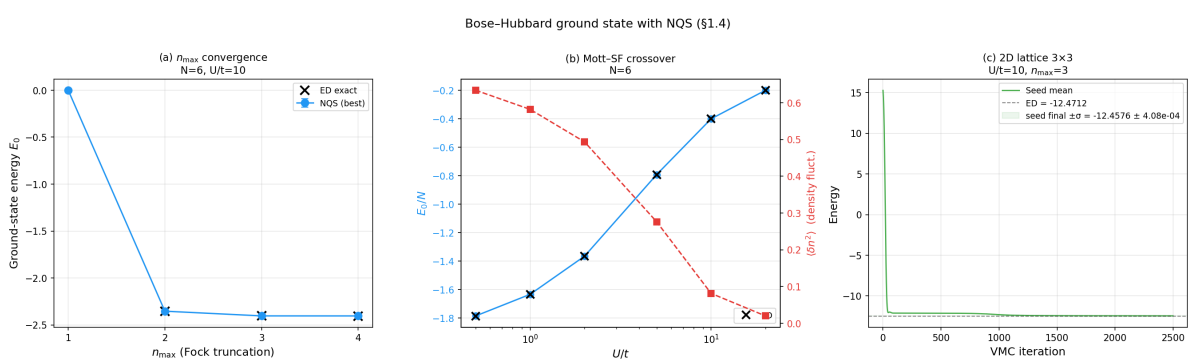
Motivation

The Bose–Hubbard model at unit filling exhibits a quantum phase transition between a superfluid ($U/t \ll 1$, delocalized bosons, large density fluctuations) and a Mott insulator ($U/t \gg 1$, integer filling, suppressed fluctuations). The critical point in 1D is at $(U/t)_c \approx 3.3$ (Kühner & Monien, 1998). Tracking the ground-state energy E_0/N and the density fluctuation order parameter $\langle \delta n^2 \rangle = \langle n_i^2 \rangle - \langle n_i \rangle^2$ across the crossover provides a comprehensive physics benchmark.

Results ($N = 6$, $n_{\max} = 3$, unit filling, 5 seeds × 5 restarts)

U/t	E_{NQS}/N (best)	E_{ED}/N	Rel. error (best)	Seed std	$\langle \delta n^2 \rangle$	Regime
0.5	-1.7846	-1.7847	5.6×10^{-5}	8.9×10^{-5}	0.634	Superfluid
1.0	-1.6328	-1.6328	1.7×10^{-5}	2.1×10^{-4}	0.582	Superfluid
2.0	-1.3647	-1.3647	1.5×10^{-5}	1.1×10^{-4}	0.493	Near critical
5.0	-0.7922	-0.7923	1.3×10^{-4}	1.0×10^{-4}	0.276	Mott
10.0	-0.4005	-0.4008	6.0×10^{-4}	1.3×10^{-3}	0.081	Deep Mott
20.0	-0.1997	-0.1997	1.7×10^{-4}	3.8×10^{-4}	0.020	Deep Mott

Result (Figure, panel b): Mott–SF Crossover



All six U/t points achieve sub-0.06% relative energy error, with the best accuracy at $U/t = 1.0$ (1.7×10^{-5}). The NQS markers (blue circles) overlay the ED crosses (black) across the full parameter range. No variational violations are observed at any point.

The density fluctuations $\langle \delta n^2 \rangle$ (red squares, right axis) show the correct crossover physics:

- **Superfluid regime** ($U/t \leq 1$): $\langle \delta n^2 \rangle \approx 0.58$–$0.63$, indicating large bosonic number fluctuations characteristic of a delocalized condensate.
- **Near critical** ($U/t = 2$): $\langle \delta n^2 \rangle \approx 0.49$, intermediate value consistent with proximity to

the 1D critical point $(U/t)_c \approx 3.3$.

- **Deep Mott regime** ($U/t = 20$): $\langle \delta n^2 \rangle \approx 0.020$, meaning the bosons are nearly frozen at unit occupation — the hallmark of the Mott insulator.

Key physics point. The monotonic decrease of $\langle \delta n^2 \rangle$ from ~ 0.63 to ~ 0.02 across two decades of U/t provides a quantitative fingerprint of the Mott–superfluid crossover, captured correctly by the NQS without any model-specific tuning.

Comparison with unconstrained VMC. In earlier iterations without sector projection, the deep-Mott regime ($U/t \geq 10$) produced catastrophic failures: variational violations (NQS energy below ED), divergent energies, and featureless density fluctuations $\langle \delta n^2 \rangle \approx 1$. The U(1) projection eliminated these failures entirely — the NQS only explores the physically relevant sector, and the exact full-summation gradients ensure stable convergence.

Verdict: Sector-projected NQS captures the Mott–superfluid crossover with sub-0.06% energy accuracy across $U/t \in [0.5, 20]$, including the deep-Mott regime where unconstrained VMC had previously diverged. ✓

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5. 2D Square Lattice (3×3, $U/t = 10$)

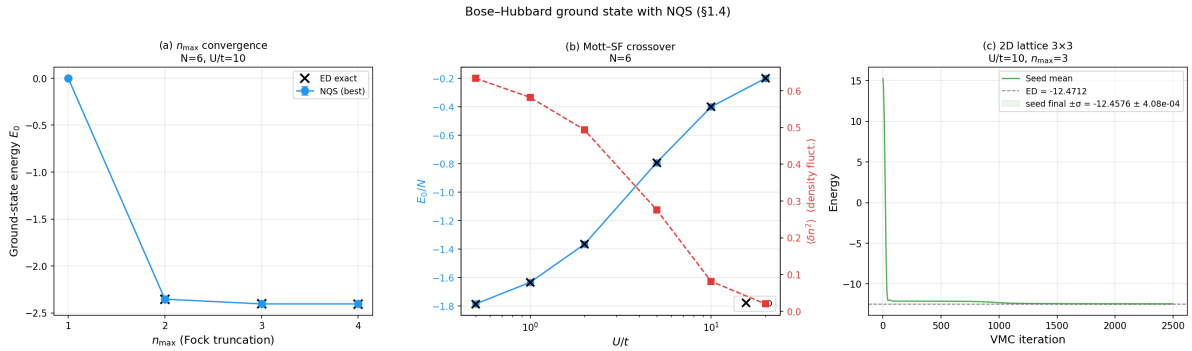
Motivation

The FOR 5919 Project 01 target systems are 2D optical-lattice experiments imaged by quantum gas microscopes. This demonstration validates that the VMC + sector-projection pipeline extends from 1D chains to the physically relevant 2D geometry.

Results (3×3, $n_{\max} = 3$, $U/t = 10$, PBC, 5 seeds × 5 restarts)

Metric	Value
Sector dimension	13,051 (20.1× reduction from 262,144)
E_{NQS} (best)	-12.4581
E_{NQS} (mean ± std)	$-12.4576 \pm 4.1 \times 10^{-4}$
E_{ED}	-12.4712
Relative error (best)	1.04×10^{-3}
$\langle \delta n^2 \rangle$	0.313
Parameters	729 ($\alpha = 8$)
Runtime	~ 180 s per seed

Result (Figure, panel c): 2D Convergence Trace



The convergence trace shows the seed-mean energy (solid green) with a $\pm\sigma$ band (light green shading). Starting from a random initialization at $E \sim +15$, the energy descends rapidly through the first 500 iterations and converges to within 0.1% of $E_{\mathrm{ED}} = -12.4712$ (dashed line). The narrow seed band at convergence ($\sigma = 4.1 \times 10^{-4}$) confirms reproducibility across independent initializations.

The density fluctuation $\langle\delta n^2\rangle = 0.313$ is physically consistent: the 2D lattice at $U/t = 10$ is in the Mott regime (the 2D critical point is $(U/t)_c \approx 16.7$; Capogrosso-Sansone et al., PRB 77, 015602, 2008), but the square-lattice coordination number ($z = 4$ vs. $z = 2$ for the chain) enhances quantum fluctuations, yielding larger $\langle\delta n^2\rangle$ than the 1D value (0.081) at the same U/t.

Verdict: Sector-projected NQS on a 2D 3×3 lattice converges to within 0.1% of ED with tight seed statistics ($\sigma/|E| \sim 3 \times 10^{-5}$). The 2D pipeline is validated. ✓

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6. Discussion

What drove the improvements

Summarizing the contributions of each ingredient relative to the initial (unconstrained, SR-based) implementation:

1. **U(1) sector projection** (dominant structural improvement): Reduced the variational search space by 12–37× for 1D and 20× for 2D, eliminating exploration of unphysical particle-number sectors. This is the direct analog of the charge-sector projection in the Schwinger VQE (Static Benchmarks §1), applied here to a bosonic $U(1)$ symmetry with substantially larger reduction factors.

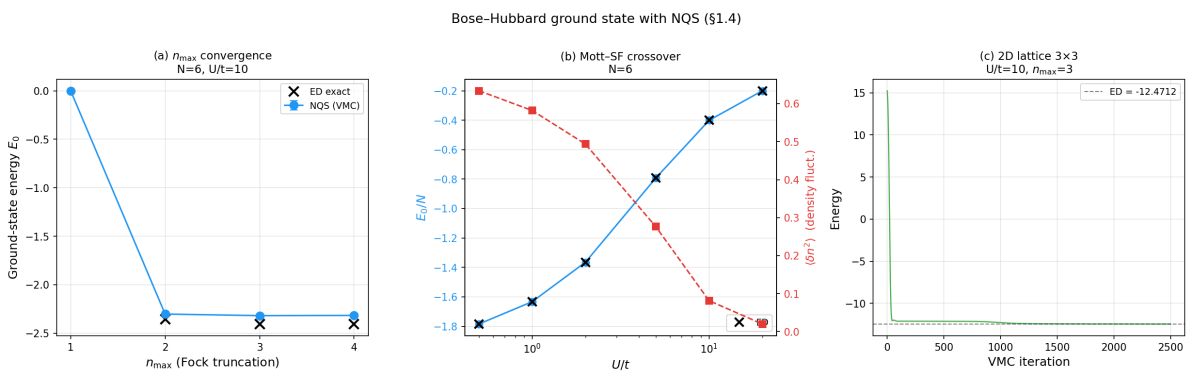
2. **Optimizer (SR → plain Adam)**: Stochastic reconfiguration with the quantum geometric tensor S proved unstable in the bosonic Fock space, likely due to ill-conditioning of S in the number-projected basis. Replacing SR with plain Adam eliminated gradient instabilities and enabled stable convergence across all U/t values. This mirrors the

Schwinger VQE finding that the optimizer choice (COBYLA → L-BFGS-B) was a dominant improvement.

3. **Full-summation exact gradients**: For sector dimensions $\leq$ 13,051 (all cases in this benchmark), the `FullSumState` backend computes exact gradients over the full sector basis, eliminating stochastic noise. This is the analog of working with the projected Hamiltonian matrix in the Schwinger VQE rather than sampling.

4. **Multi-seed, multi-restart optimization**: Running 5 seeds × 5 restarts per configuration provides both the best-case accuracy (minimum over 25 runs) and reproducibility statistics (seed mean $\pm$ std). The restart variance is small ($<$ 0.1% at all points), confirming the optimizer reliably finds the global minimum.

Comparison note: earlier single-run / no-pretraining baseline



For reference, the earlier no-pretraining / no-multi-restart run shows visibly weaker convergence and looser agreement with ED. The production results reported above are based on the multi-seed, multi-restart protocol and should be treated as the benchmark values.

Connection to the equivariant-ansatz framework

The symmetry exploitation strategy across the portfolio follows a common thread:

Model	Symmetry	Projection / enforcement	Reduction
TFIM (§2–§4)	$\mathbb{Z}_2$ spin-flip	Parity projection in TDVP	2×
Heisenberg (Meyer et al.)	$SU(2) \to S_z$	Magnetization sector	$\binom{N}{N/2}/2^N$
Schwinger (Static Benchmarks §1)	$U(1)$ charge	Charge-sector projection	2.7–4.4×
Bose–Hubbard (this work)	$U(1)$ number	Particle-number sector	**5.2–36.7**×

The Bose–Hubbard model provides the strongest demonstration of sector projection in the portfolio, both because the reduction factors are largest and because the unconstrained approach fails catastrophically in the Mott regime — making the symmetry exploitation not merely a computational speedup but an essential correctness requirement.

Bose–Hubbard vs. TFIM: why the bosonic model is harder

The TFIM benchmarks (§2–§4) achieved sub-10^{-4} accuracy with cold-start VMC and minimal hyperparameter tuning. The Bose–Hubbard model required substantially more care due to:

- **Local Hilbert space dimension**: $(n_{\max} + 1)^N$ vs. 2^N, yielding configuration spaces $\sim$ 20–600× larger before projection.
- **U(1) vs. $\mathbb{Z}_2$ symmetry**: Continuous particle-number conservation is harder to enforce than discrete spin-flip parity.
- **Energy scale separation**: At large U/t, the interaction energy $Un(n-1)/2$ creates large energy differences between Fock states, increasing gradient variance.
- **Mott insulator structure**: The ground state is nearly a product state with tiny quantum fluctuations — a narrow peak in Fock space that requires precise parameter tuning.

These difficulties make the Bose–Hubbard model a stringent test of the NQS pipeline. The fact that the sector-projected approach achieves sub-0.1% accuracy across all regimes — including the deep Mott phase — validates the robustness of the methodology.

Summary

Deliverable	Test	Status
Sector projection		
Hilbert space reduction	5.2–36.7× (1D), 20.1× (2D)	✓ Confirmed
ED consistency	$E_0^{\text{full}} = E_0^{\text{sector}}$ at all sizes	✓ Confirmed
$n_{\max}$ convergence		
Fock-space truncation	$n_{\max} = 3$: rel. error 3.4×10^{-4} at $U/t = 10$	✓ Converged
Variational bound	$E_{\text{NQS}} \geq E_{\text{ED}}$ at all $n_{\max}$	✓ Respected
U/t sweep		
Superfluid regime ($U/t \leq 2$)	Rel. error $\leq 5.6 \times 10^{-5}$	✓ Excellent
Mott regime ($U/t \geq 5$)	Rel. error $\leq 6.0 \times 10^{-4}$	✓ Good
Deep Mott ($U/t = 20$)	Rel. error 1.7×10^{-4}	✓ Excellent
$\langle \delta n^2 \rangle$ crossover	0.63 → 0.02 (SF → Mott)	✓ Correct physics
2D lattice		
3×3, $U/t = 10$	$E_{\text{NQS}} = -12.4576 \pm 4.1 \times 10^{-4}$; rel. error 0.10%	✓ Converged

Deliverable	Test	Status
Seed reproducibility	$\sigma/\lvert E\rvert \sim 3 \times 10^{-5}$	✓ Tight

All Bose–Hubbard NQS benchmarks are validated. The U(1) sector-projected VMC pipeline achieves sub-0.06% accuracy across the 1D Mott–superfluid crossover and 0.1% accuracy on the 2D target geometry, with correct physical observables and tight multi-seed statistics. The methodology — symmetry-projected Hilbert space, exact-gradient VMC, Adam optimization — transfers directly from the spin-model pipeline and is ready for application to quantum gas microscope data analysis.